

How gamma controls readout learning

An attainable target can still be expensive to learn. With identical centers, target, and zero initialization, frozen tanh readout reaches 1% relative error after **15.8 million GD updates at gamma 8**, versus **16,013 at gamma 64**. Adam also misses 1% at gamma 8 within 200,000 updates. The mechanism is explicit: gamma smooths features, smoothing changes correlations with needed error patterns, and the target-weighted spectrum determines GD speed.

Notation. Eigenvalues use μ ; norms are Euclidean.

$J_\gamma, K_\gamma = J_\gamma J_\gamma^T$	Normalized features and readout kernel
$\mu_i, u_i; \rho_i = \mu_i / \mu_1$	Eigenpairs; eigenvalue relative to largest
$y, r_n; p_i = u_i^T y ^2 / \ y\ ^2$	Target, residual; target energy in mode i

1 Why feature correlations control learning

Train only readout w in $J_\gamma w$ on m input samples x_a , starting at zero. The W tanh features share slope γ and consecutive equally spaced centers c_j , spacing h :

$$J_\gamma[a, j] = \frac{\tanh(\gamma(x_a - c_j))}{\sqrt{m}}, \quad J_\gamma[a, 0] = \frac{1}{\sqrt{m}}, \quad y_a = \frac{f(x_a)}{\sqrt{m}}.$$

Minimizing $\frac{1}{2} \|J_\gamma w - y\|^2$ is half mean-squared error. GD gives $r_{n+1} = (I - \eta K_\gamma) r_n$. For a unit error pattern v ,

$$q_\gamma(v) = v^T K_\gamma v = \|J_\gamma^T v\|^2 = \sum_i \mu_i(\gamma) |u_i(\gamma)^T v|^2.$$

This is total squared correlation with the features: an eigenvector's learning eigenvalue, or a weighted average for other patterns. If the residual is v , the correction along v is $\eta q_\gamma(v)$. The largest eigenvalue limits the shared stable step, $0 < \eta < 2/\mu_1$, making relative eigenvalues ρ_i central to learning speed.

2 The explicit gamma mechanism

A tanh is a smoothed step: $\tanh(\gamma \cdot) = s_\gamma * \text{sign}$ with $s_\gamma(x) = \frac{\gamma}{2} \text{sech}^2(\gamma x)$. Its Fourier multiplier is

$$M_\gamma(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}, \quad M_\gamma(0) = 1.$$

Small gamma averages more widely and suppresses rapid variation. Replace the center sum by an integral: subtracting the whole-line deficit of two tanh features from their saturated-product baseline gives B_γ . The correction R_γ restores finite endpoints and discrete centers:

$$B_\gamma[a, b] = \frac{W+1}{m} - \frac{2}{hm} d_{ab} \coth(\gamma d_{ab}), \quad d_{ab} = x_a - x_b, \quad R_\gamma = K_\gamma - B_\gamma.$$

At $d = 0$, use $d \coth(\gamma d) = 1/\gamma$.

Theorem 1 (Gamma-dependent kernel response). For these centers, fixed samples, $\Gamma > \gamma > 0$, and any real v , set $V_v(\omega) = \sum_a v_a e^{-i\omega x_a}$. Then

$$q_\Gamma(v) - q_\gamma(v) = \mathcal{A}_{\Gamma, \gamma}(v) + v^T (R_\Gamma - R_\gamma) v,$$

$$\mathcal{A}_{\Gamma, \gamma}(v) = \frac{2}{\pi h m} \int_{\mathbb{R}} \frac{M_\Gamma(\omega)^2 - M_\gamma(\omega)^2}{\omega^2} |V_v(\omega)|^2 d\omega \geq 0.$$

In words: response changes by explicit gain from reduced smoothing plus a finite-geometry correction. The pattern enters through $|V_v|^2$; gamma enters through $M_\Gamma^2 - M_\gamma^2$. Pairing two features gives the squared multiplier. The identity is exact, including the continuous zero-frequency limit. The signed correction need not be small in every direction. Appendix A gives the proof.

Direct validation. Use $W = 559$, $h = 1/256$, $m = 8,193$ uniform samples on $[-1, 1]$, and $f(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x)$. At gamma 8, the most target-aligned resolved mode with $0 < \rho_i \leq 2 \times 10^{-6}$ is mode 22, carrying **4.14% of target energy**. This retrospective choice uses the initial kernel and target, without training outcomes. Freeze v and predict larger-gamma response by $q_8(v) + \mathcal{A}_{\Gamma,8}(v)$. Figure 1 shows roughly **1,000-fold growth**, with discrepancy below **0.061%** at gammas 12, 16, and 64. These corrections are measured for this probe. The gain uses equivalent closed-form entries $2[d \coth(8d) - d \coth(\Gamma d)]/(hm)$, with no new eigenspace calculation.

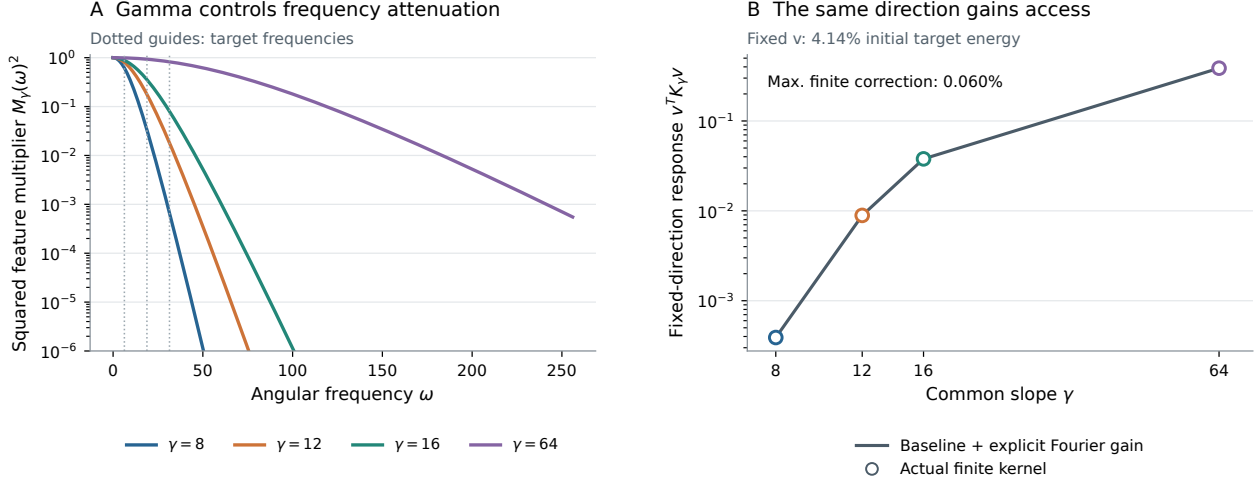


Figure 1: The mechanism predicts the response change. Left: squared multipliers; dotted guides mark target frequencies. Right: direct finite-kernel response (circles) and reference plus Fourier gain (line). The same gamma-8 eigenvector is used throughout; it need not remain an eigenvector at larger gamma.

3 From the mechanism to measured learning

A larger average response can coexist with slow components. To determine acquisition time, retain each eigenvalue and target weight. From zero, classical GD gives

$$e(n)^2 := \frac{\|r_n\|^2}{\|y\|^2} = \sum_i p_i (1 - \eta \mu_i)^{2n} = \sum_i p_i (1 - \rho_i/2)^{2n} \quad \text{if } \eta = 1/(2\mu_1).$$

Nullspace factors are one. A ratio 10^{-6} needs about two million updates for one e-fold reduction. Measured finite spectra, target weights, and actual archived steps predict **15,798,313; 186,057; 61,792; and 16,013 updates** to 1% at gammas 8, 12, 16, and 64, exactly matching executed crossings. All four attain the studied accuracy; no decay rate is fitted to training.

Adam tests whether the same difficult directions retain error. Adaptive updates do not follow GD’s eigenmode law. In Figure 2B, the gamma-8 band $0 < \rho_i \leq 2 \times 10^{-6}$ starts with **4.33%** target energy and retains 1.47094×10^{-4} after 200,000 Adam updates, above the 10^{-4} squared-error budget. At larger gamma, band energies are below budget. Normalization uses $\|y\|^2$; ratio-defined subspaces can change with gamma.

Figure 2C shows first crossings of the EMA of squared relative error with one shared **100-update half-life**: no crossing at gamma 8, then **7,688; 1,900; and 1,486 updates**. Recrossings occur: this is first acquisition, not sustained accuracy. Initial-loss memory prevents crossing before **1,329 updates**, compressing the fastest-run differences. Appendix B gives loss curves, protocol, and sensitivity checks.

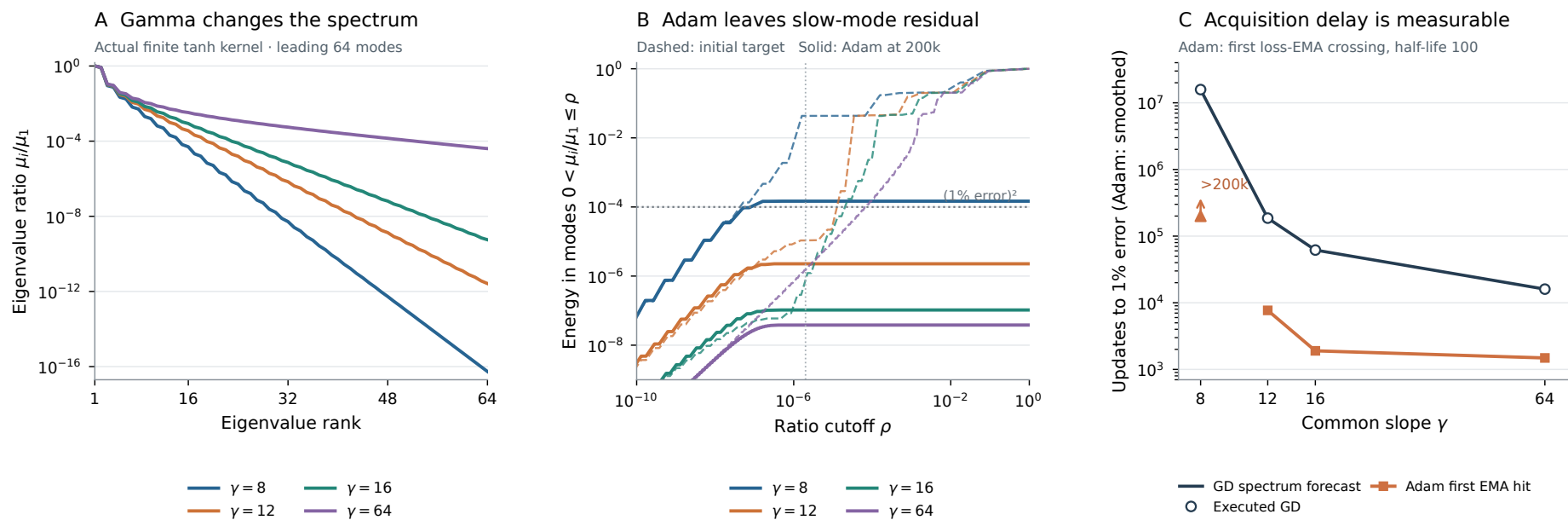


Figure 2: Spectrum, target overlap, and acquisition. A: leading 64 finite-kernel eigenvalue ratios, ordered separately at each gamma. B: cumulative initial target energy (dashed) and Adam residual energy at 200,000 (solid). The blue residual exceeds the 1% budget in slow modes alone. C: executed GD crossings and measured-spectrum forecasts, alongside first Adam EMA crossings. The arrow means censoring at 200,000; GD has a longer budget. All panels use the same target and geometry.

A Full proof

Let $I = [c_1 - h/2, c_W + h/2]$ and $F_{ab,\gamma}(c) = \tanh(\gamma(x_a - c)) \tanh(\gamma(x_b - c))$. The identity $1 - \tanh u \tanh v = \coth(u - v)(\tanh u - \tanh v)$ and the integral of translated tanh profiles give

$$\int_{\mathbb{R}} (1 - F_{ab,\gamma}(c)) dc = 2d_{ab} \coth(\gamma d_{ab}).$$

The coincident limit is $2/\gamma$. Since $|I| = Wh$, subtract this deficit from the constant baseline and restore the exterior part to get $K_\gamma = B_\gamma + R_\gamma$, where

$$R_\gamma[a, b] = \frac{1}{m} \left[\sum_{j=1}^W F_{ab,\gamma}(c_j) - \frac{1}{h} \int_I F_{ab,\gamma}(c) dc \right] + \frac{1}{hm} \int_{\mathbb{R} \setminus I} (1 - F_{ab,\gamma}(c)) dc.$$

These are the quadrature discrepancy and finite-interval correction. Bias contributes $1/m$ to B_γ and cancels across slopes. Both $s_\gamma * \text{sign}$ and $\tanh(\gamma \cdot)$ vanish at zero and have derivative $2s_\gamma$, proving the smoothing identity. With $\widehat{g}(\omega) = \int g(x) e^{-i\omega x} dx$, substitute $u = e^{2\gamma x}$ and $a = \omega/(2\gamma)$:

$$\begin{aligned} \widehat{s}_\gamma(\omega) &= \int_0^\infty \frac{u^{-ia}}{(1+u)^2} du = \Gamma(1-ia)\Gamma(1+ia) \\ &= \frac{\pi a}{\sinh(\pi a)} = M_\gamma(\omega). \end{aligned}$$

The beta integral and gamma-function reflection identity give the middle equalities; $\Gamma(\cdot)$ here denotes the gamma function. Put $t_\gamma(u) = \tanh(\gamma u)$. Differentiate $\int [1 - t_\gamma(u)t_\gamma(u-d)] du$ twice and integrate by parts:

$$\begin{aligned} \frac{d^2}{dd^2} [2d \coth(\gamma d)] &= - \int t_\gamma(u) t_\gamma''(u-d) du \\ &= \int t_\gamma'(u) t_\gamma'(u-d) du = 4(s_\gamma * s_\gamma)(d). \end{aligned}$$

The boundary term vanishes and s_γ is even. Thus the integrable difference $D(d) = 2[d \coth(\gamma d) - d \coth(\Gamma d)]$ has transform

$$\widehat{D}(\omega) = \frac{4(M_\Gamma(\omega)^2 - M_\gamma(\omega)^2)}{\omega^2}, \quad \widehat{D}(0) = \frac{\pi^2}{3}(\gamma^{-2} - \Gamma^{-2}).$$

Use $\widehat{D}'' = -\omega^2 \widehat{D}$ and expand $z/\sinh z$ at zero. This multiplier decreases for $z > 0$, so $M_\Gamma \geq M_\gamma$ and $\widehat{D} \geq 0$. Finally, $B_\Gamma[a, b] - B_\gamma[a, b] = D(x_a - x_b)/(hm)$. Fourier inversion contributes $1/(2\pi)$; summing against $v_a v_b$ gives $|V_v|^2$ and prefactor $2/(\pi hm)$. Add the exact finite correction to complete the proof. $\square$

B Adam protocol and supporting checks

Metric and protocol. Common Adam uses initial rate 10^{-3} , epsilon 10^{-12} , and moments $(0.9, 0.999)$. Rate stays fixed through 20,000, decays by cosine to 10^{-6} at 50,000, then stays fixed through 200,000. Define

$$M_0 = 1, \quad M_n = \beta M_{n-1} + (1 - \beta)e(n)^2, \quad \beta = 2^{-1/100}.$$

First EMA acquisition is the first $M_n \leq 10^{-4}$, calculated from every update; plots show $\sqrt{M_n}$. Later upcrossings occur 73, 97, and 98 times at gammas 12, 16, and 64; raw sustained crossings occur around 41,000. Since $M_n \geq \beta^n$, memory imposes the 1,329-update floor. The shared window is a retrospective visualization choice.

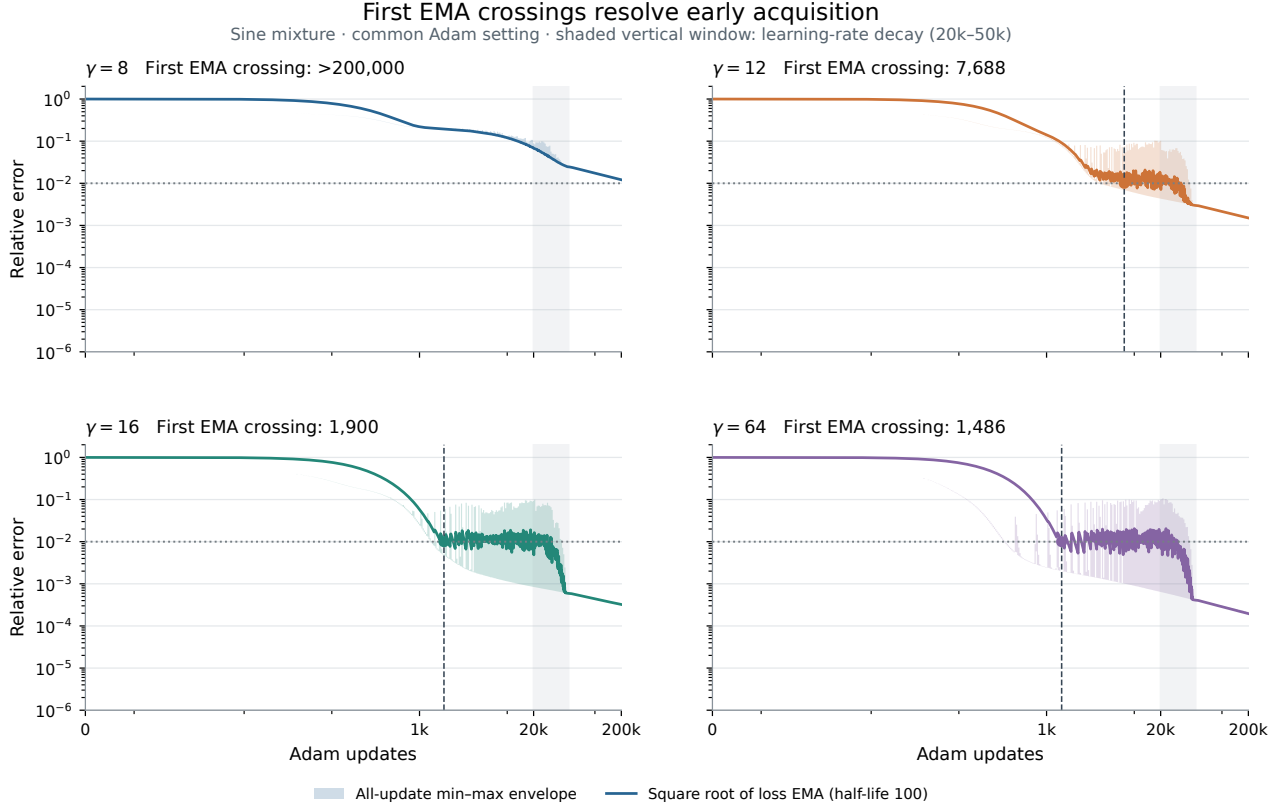


Figure B1: Actual training and the acquisition metric. Bands show raw-error minima/maxima within bins; lines show the loss EMA's square root. Dashed vertical lines mark first crossings; gray windows mark rate decay. Curves are sampled; crossings use every update.

Robustness. Targets are the sine mixture, $\exp(\sin(3\pi x))$, $1/(1 + 25x^2)$, $\sqrt{5}x^2$, and $\sqrt{2}\sin(2\pi x)$. Besides the common setting, archived settings were selected from five rates and two epsilons by median validation error at five checkpoints from 40,000 to 50,000; continuation preserved optimizer state. Selection rewards late accuracy, not early acquisition. Figure B2 shows target and optimizer dependence.

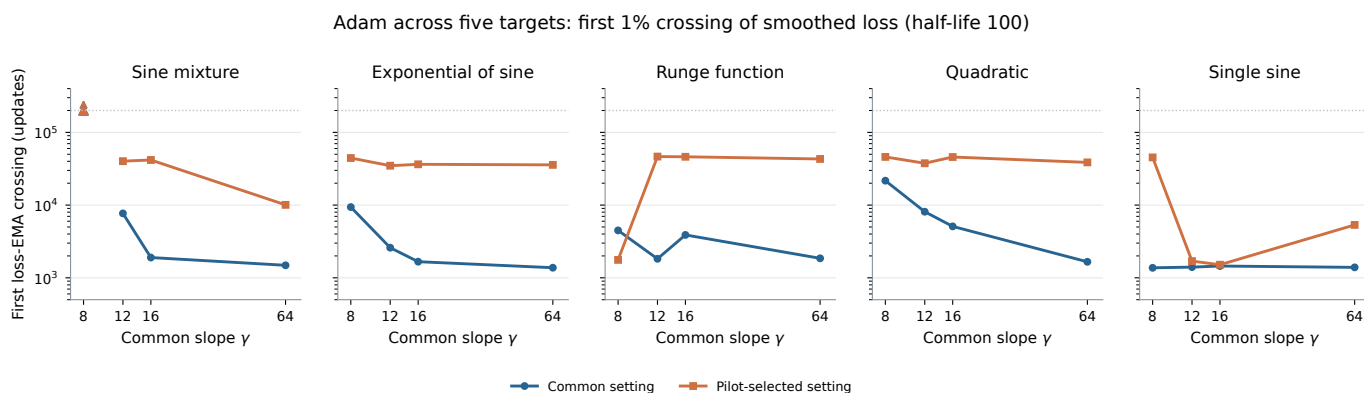


Figure B2: Target and optimizer dependence. All cases use the 100-update half-life. Arrows mean no crossing by 200,000. Results do not support universal monotone improvement with gamma.

The primary common-setting ordering persists at half-lives 30, 100, and 300. At 1,000, crossings cluster near 28,000–30,000 and reorder. Validation-selected settings also change the comparison (Figure B3).

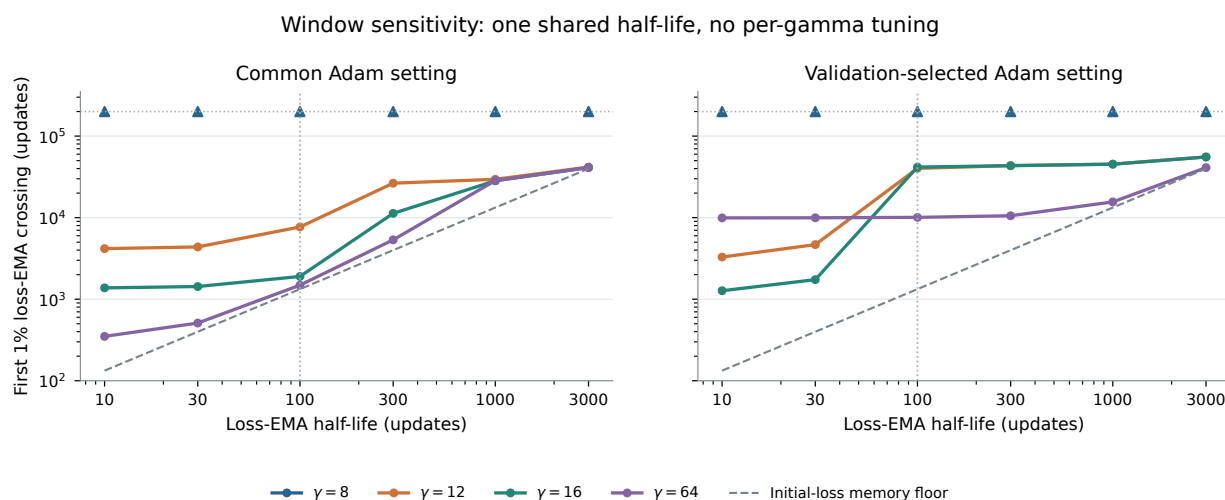


Figure B3: Window sensitivity. Each horizontal position uses one half-life across gammas. Dotted lines mark 100; the gray dashed line is initial-loss memory. Triangles mean censoring at 200,000. All points use the same saved trajectories.

Numerical evidence. Archived FP64 calculations retain singular values above 10^{-14} of the largest; omitted directions are tracked separately from exact nullspace. Independent SVD and residual projections reproduce slow-band energies within 5.1×10^{-14} . Adam projections measure current error and may increase during training. These are training-grid optimization results. The [evidence directory](#) contains plotted values, hashes, and verification; no new training was run.

Reproduce from the repository root:

```
export OPENBLAS_NUM_THREADS=1 VECLIB_MAXIMUM_THREADS=1
python -m experiments.expD36_frozen_gamma_probe.gamma_mechanism_analysis
python -m experiments.expD36_frozen_gamma_probe.adam_spectral_analysis
python -m experiments.expD36_frozen_gamma_probe.adam_ema_analysis \
    --archive /path/to/adam_raw_scalar_archive.npz
python -m experiments.expD36_frozen_gamma_probe.optimizer_access_figure
latexmk -pdf -outdir=/tmp/gamma-optimizer-latex docs/gamma_optimizer_access_note.tex
```

Extract scalar traces with `adam_trace_audit.py`; retained compact EMA artifacts suffice to regenerate figures without extraction.